

SOCIAL INTERACTION TRACKER

INTRODUCTION:-

A SOCIAL INTERACTION TRACKER SYSTEM USES BLE (BLUETOOTH LOW ENERGY) PROXIMITY SENSORS IN WEARABLE DEVICES TO DETECT NEARBY DEVICES. IT MEASURES SIGNAL STRENGTH (RSSI) TO ESTIMATE DISTANCE AND IDENTIFY INTERACTION LEVELS BETWEEN USERS.

OBJECTIVES:-

- TO DETECT NEARBY DEVICES USING BLE TECHNOLOGY
- TO MEASURE SIGNAL STRENGTH (RSSI)
- TO CLASSIFY INTERACTION STRENGTH (STRONG / MEDIUM / WEAK)
- TO STORE INTERACTION DATA IN ARRAYS
- TO ANALYZE USER INTERACTION PATTERNS
- TO BUILD A SIMPLE AND EFFICIENT TRACKING SYSTEM

TECHNOLOGY USED:-

- BLE (BLUETOOTH LOW ENERGY)
- RSSI (RECEIVED SIGNAL STRENGTH INDICATOR)
- WEARABLE DEVICES
- PYTHON / EMBEDDED SYSTEMS
- DATA STRUCTURES (ARRAYS)

IMPLEMENTATION:-

```
1 import random
2 # Simulated nearby devices with RSSI values
3 devices =
4     'Device_A = random.randint(-90, -30), -30)
5     'Device_B = random.randint(-90, -30), -30)
6     'Device_C = random.randint(-90, -30), -30)
7
8 # Function to classify interaction
9 def classify(rssi):
10     if rssi > -50:
11         return 'Strong'
12     else:
13         return 'Weak'
14
15 # Store interaction data
16 for device, rssi in devices.items():
17     strength = classify(rssi)
18     interaction_data.append((device, rssi), strength)
19
20 # Output results
21 for data in interaction_data:
22     print('Device: ' + data[0] + ' | RSSI: ' + str(data[1]) + ' | Interaction: ' + data[2])
```

FLOWCHART:-



OUTCOME:-

THE SYSTEM SUCCESSFULLY TRACKS NEARBY INTERACTIONS USING BLE SIGNALS.

IT CLASSIFIES INTERACTION STRENGTH BASED ON RSSI VALUES AND STORES THE DATA EFFICIENTLY.

THIS PROJECT DEMONSTRATES A REAL-WORLD IOT APPLICATION FOR SOCIAL BEHAVIOR ANALYSIS.